`bitop/v1` completeness of the symmetrised oracle

`research/ic_acct7_symmetrised_20261021` · session 2026-10-21 · report 2026-10-26, repriced 2026-10-27 · class: accounting

Question

`PROTOCOL.md` section A (frozen 2026-10-21, before any data): after commit `fd93ddfe2` adds counters for the `koblitz-symmetrised` base build, the symmetrised oracle's `prepare`, the symmetrised system build and the degree scan, **does every `ic.pipeline_full` record with `oracle=symmetrised` price `complete: true` (nothing unpriced, nothing known-uncounted) under `ecbench work --counted`?** Pass requires every counted record of the IC arm complete and a verify receipt with zero problems.

This is an accounting change: it makes records honest about their full cost. It is not an improvement of the index-calculus baseline and makes no security or speed claim. Section B (the Koblitz exponent ladder, `specs/koblitz-exponent-n{13,19,23}.json`) is frozen and **not run**.

Session

- **Spec** `specs/koblitz-symmetrised-n17.json` → `sessions/n17-symmetrised` (`ECBS1h6fd757ff9d9f`). Curve $K_1/\mathbb{F}_{2^{17}}$ (`icv1-f2m17-tm101-00378d4e`), 4 public targets, 2 measured rounds plus 1 warm-up, alternate order, `ecbench run --cpus 1` (L0), binary sha256 `bf8196c2…`.
- **Arms.** `ic-symmetrised-d9` (`ic.pipeline_full`, `koblitz-symmetrised:divisor=0;1`, `symmetrised:m=2,divisor=0;1`, `targets=walk`) and the reference `rho-frob` (`rho.signed_frobenius_full`).
- **Runs.** 24 records (8 warm-up), 24 verified; 16 counted (8 per arm).
- **Audit.** `ecbench verify --replay 2`: `audit OK: 24 records, 24 verified, 2 replays reproduced, 0 problems` (`audit/n17-verify-replay2.json`).

Result (section A): pass

Every counted record of both arms has `complete: true`, `unpriced: []` and `known_uncounted: []` (`audit/n17-work-counted.jsonl`).

Correction (review of PR #2638). As first priced, the `koblitz-symmetrised` base's Artin–Schreier lifts and orbit walks were charged twice: in the builder's `tally_*` counters and again through the natives `as_solves` (511) and `frobenius_maps` (409) that `koblitz_factor_base` derives from the base's size. The natives stay in the record for replay and are now priced once (a `tallied:<native>` marker; a `koblitz-symmetrised` base phase carrying a tally is read as marked, which covers this session's records). The figures below are re-derived from the committed session with the fixed binary (sha256 `4ff2214e…`): each IC record falls by exactly 2,470,915 gates, all in `factor_base`; every rho record is unchanged. Medians are taken by `audit/medians.jq` (`audit/n17-work-medians.json`).

arm	counted	complete	median bitop/v1 (min–max)	median GAE
`ic-symmetrised-d9`	8	8 / 8	1.956e9 (0.980–3.664e9)	897.1
`rho-frob` (reference)	8	8 / 8	1.868e6 (1.269–6.582e6)	254.6

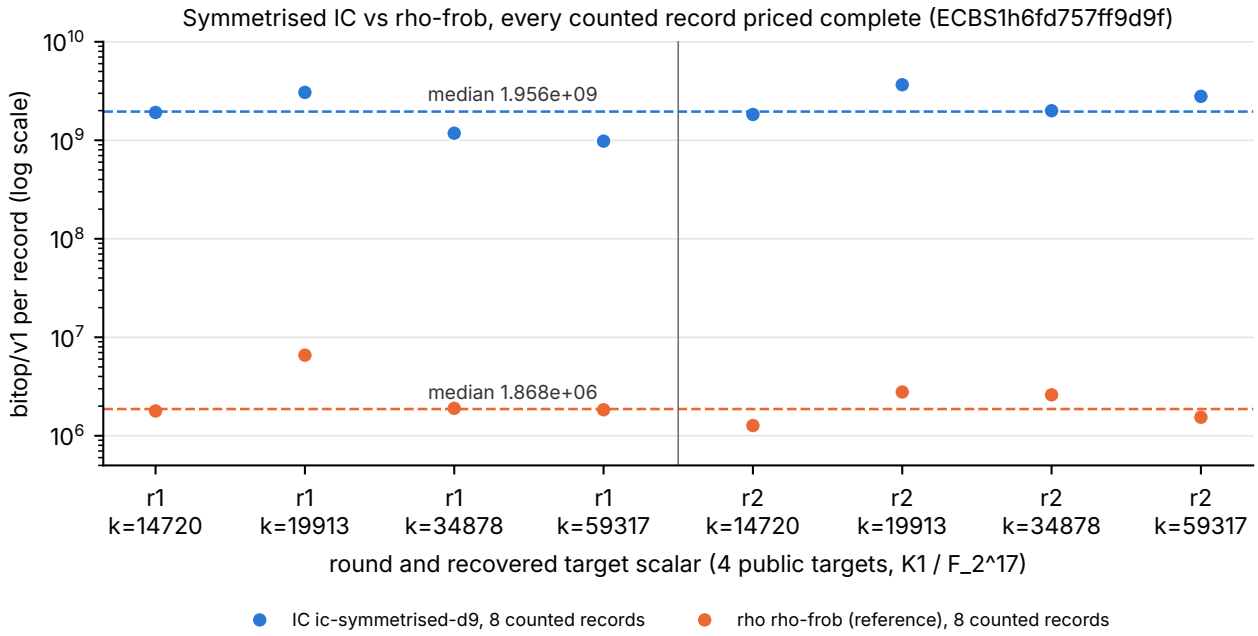


Figure 1: `bitop/v1` per counted record, both arms, by round and recovered target scalar; dashed lines are each arm's median of 8 (the README's recorded medians, asserted by `figures/plots.py`). Session `ECBS1h6fd757ff9d9f`; exact counts, no interval stated (4 targets at 17 bits support none).

Distance remaining. As the protocol requires, no speed claim is drawn from four 17-bit targets. Stated as the distance remaining, the median IC record costs about 1047× the matched rho in `bitop/v1` (1.956e9 against 1.868e6; first priced 1048×). One IC record (seq 8) splits by phase as `relations` 1.897e9, `factor_base` 1.19e7 (first priced 1.44e7), `oracle_setup` 6.86e6, `linear_algebra` 4.9e5 and `verify` 7.3e4.

Instructions per counted gate (`ecbench audit-ir`)

`ecbench audit-ir --spec specs/koblitz-symmetrised-n17.json --seq 8 9 10 11` re-executed four plan entries under Callgrind (`--cache-sim=no --branch-sim=no`) with the session's binary; each reproduced its record's bitop count exactly (`audit/n17-audit-ir.jsonl`, with the first pricing's gate counts). The Ir is unchanged by the repricing; `audit/n17-audit-ir-repriced.jsonl` divides each recorded `solve_ir` by the repriced gate count (jq, `audit/audit-ir-reprice.jq`), without re-running Callgrind.

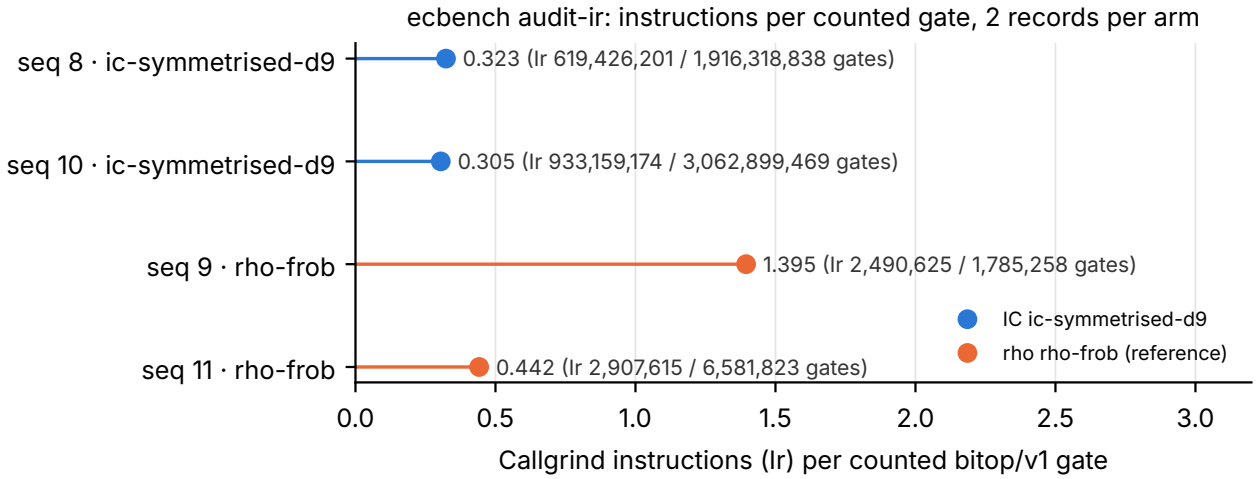


Figure 2: Callgrind solve instructions per counted `bitop/v1` gate (the repriced `ir_per_kilo_bitop_repriced` / 1000), two records per arm.

The symmetrised arm executes 0.323 and 0.305 instructions per counted gate (first priced 0.323 and 0.304); the rho reference 1.395 and 0.442. The rho solves are short (about 2–7e6 gates), and per-solve work that does not grow with

the walk would produce that spread, but this has not been checked against the Callgrind parts. Two records per arm describe this session and support no general ratio.

Limitations

- **Section B not run.** The exponent ladder (`n = 13, 19, 23` at a fixed base size) is frozen in `PROTOCOL.md` and its specs, with no sessions.
- **One small size:** $K_1/\mathbb{F}_{2^{17}}$, 4 targets, 2 rounds. No interval, no fit, no comparison claim.
- **L0 isolation** on a shared host; only counts are used, no wall time.
- **audit-ir** covers 4 of the 16 counted records.
- **Validation gate.** No `VALIDATION.json` (replay certificate, validator report), so no IC leaderboard or `benchmarks/` row. The IC/rho multiple is a diagnostic point (class accounting) on the scoreboard's progress chart.

Evidence

`PROTOCOL.md`, `specs/`, `sessions/n17-symmetrised/`, `audit/n17-verify-replay2.json`, `audit/n17-work-counted.jsonl`, `audit/n17-work-summary.jsonl`, `audit/n17-work-medians.json` (`audit/medians.jq`), `audit/n17-audit-ir.jsonl`, `audit/n17-audit-ir-repriced.jsonl` (`audit/audit-ir-reprice.jq`), `figures/plots.py` (plots recorded values only).